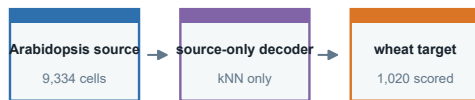


**a Zero-target labels remain physically separated**

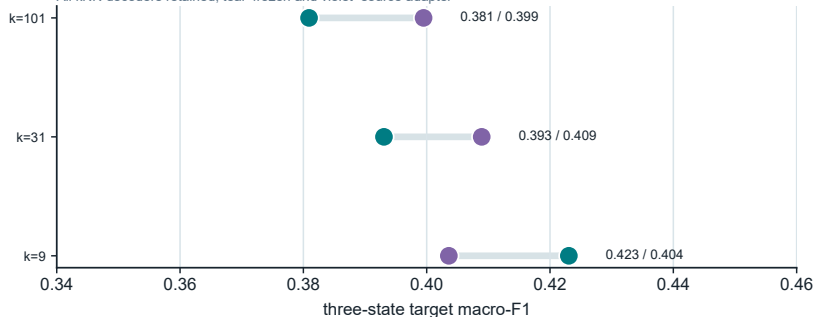
The source-only decoder is written before wheat labels are read for scoring



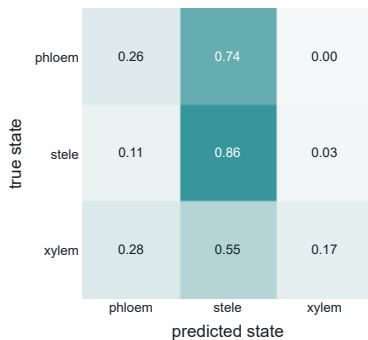
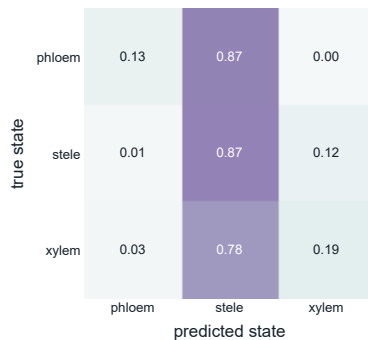
No wheat labels are used for fitting, calibration or decoder selection.

**b Source adaptation does not improve zero-target macro-F1**

All kNN decoders retained; teal=frozen and violet=source adapter

**c****Cross-species vascular-state errors remain inspectable**

Row-normalized three-state confusion; all target classes are retained

**Frozen root checkpoint (k=9)****GSE270140 source adapter (k=9)****d Why the negative result remains**

Retaining the failure prevents a selective cross-species narrative

